Design a smart-city emergency routing system with 100+ intersections (Nodes) and 200+ weighted roads (Edges and Weights)

ASSIGNMENT REPORT

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B.TECH IN COMPUTERSCIENCE

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1. Problem Statement and Problem Formulation

A smart-city traffic management company is building a real-time emergency vehicle routing system for ambulances and fire engines. The system ingests a live road network of intersections, roads, distances, and traffic conditions and must compute routes for emergency vehicles under strict operational limits.

The road network can contain more than 5,000 intersections (nodes) and 10,000 roads (weighted, and in places bidirectional, edges). The system must return a route within 200 ms, must always route vehicles along the shortest currently-available path, must support frequent live updates to road/traffic conditions, must operate within moderate memory budgets, and must serve both one-off route queries and repeated queries between the same pair of locations efficiently.

Problem Formulation: Given a weighted directed/undirected graph G = (V, E) representing intersections and roads, where edge weights represent live travel time, and given frequent updates to edge weights (traffic conditions), determine the minimum-travel-time path between a source intersection (emergency vehicle location) and a destination intersection (incident location) in near real time, while keeping repeated queries and traffic updates computationally cheap.

2. Objective and Expected Outcomes

Objectives

- Model the road network as a graph capable of holding 5,000+ nodes and 10,000+ weighted edges.
- Store the graph using hash-based adjacency lists (HashMap) for fast neighbour lookup and hash-based visited tracking (HashSet) during traversal.
- Implement Dijkstra's shortest-path algorithm using a Min-Heap priority queue for efficient extraction of the next-closest intersection.
- Support dynamic traffic updates with O(1) average-time edge-weight modification via a dedicated edge HashMap.

- Cache results of repeated route queries in a HashMap-based route cache to avoid recomputation.
- Keep the end-to-end query time within the 200 ms operational limit and memory usage within moderate bounds.

Expected Outcomes

- A working C program that builds a large-scale weighted road-network graph and answers shortest-route queries correctly.
- Demonstrated integration of Min-Heap + Dijkstra, HashMap adjacency storage, HashSet visited-tracking, and a route cache.
- Empirical timing evidence that the system meets the sub-200 ms response requirement, including after traffic updates.
- An evidence-based discussion of design trade-offs aligned with SDG 9 and SDG 11 (resilient, sustainable urban infrastructure).

3. Requirements, Constraints and Assumptions

Functional Requirements

- Support a graph of 5,000+ intersections and 10,000+ weighted roads.
- Compute the shortest (minimum travel-time) route between any source and destination intersection.
- Support live updates to individual road weights (traffic conditions) without rebuilding the whole graph.
- Support repeated one-to-one route queries efficiently using a cache.

Non-Functional Requirements

- Route computation must complete within 200 ms for the specified graph scale.
- Memory usage must remain within moderate computational resources (arrays and chained hash buckets, no dense adjacency matrix).
- The design must be modular in C: separate modules for graph storage, priority queue, hashing, and traffic updates.

Constraints

- Implementation language is C, without external graph libraries.
- Testing performed on synthetically generated road-network data (random weighted graph) rather than a live GIS feed.

Assumptions

- Road weights represent current travel time in seconds and are always non-negative, which is required for Dijkstra's algorithm to be valid.
- A traffic update changes a road's weight but never removes or adds an intersection at runtime.
- Because any traffic update can change shortest paths, the route cache is invalidated whenever an edge weight changes.

4. Application of Relevant Course Knowledge / Concepts

- Graph ADT: intersections as vertices, roads as weighted edges; adjacency-list representation chosen over adjacency matrix because the network is sparse (10,000 edges over 5,000+ nodes).
- Hashing: separate chaining hash maps used for (a) adjacency-list storage keyed by node id, and (b) edge-weight lookup keyed by the ordered pair (u, v), giving O(1) average-time neighbour and edge access.
- Hash Set: a boolean/bitmap-backed set used to mark and check visited nodes in O(1) during Dijkstra's traversal, avoiding repeated linear scans.
- Heaps / Priority Queues: a binary Min-Heap with position tracking (for O(log n) decrease-key) is used to always extract the currently-closest unvisited intersection.
- Greedy algorithm design: Dijkstra's algorithm greedily finalizes the shortest distance to the closest unvisited node at each step, which is optimal for non-negative edge weights.
- Caching / memoization: a HashMap-based route cache avoids recomputbudgeting Dijkstra's algorithm again for repeated (source, destination) queries.
- Complexity analysis: Dijkstra's algorithm with a binary heap runs in O((V + E) log V), which for V = 5,000 and E = 10,000 comfortably supports sub-200 ms responses.

5. Design / Proposed Solution / Methodology

5.1 System Modules

- Graph Storage Module – AdjHashMap: separate-chaining hash map from node id to its adjacency (neighbour, weight) list.
- Edge Lookup Module – EdgeHashMap: hash map keyed by (u, v) pair pointing directly at the corresponding adjacency-list node, enabling O(1) average traffic updates.
- Visited Tracking Module – VisitedSet: an O(1) HashSet-style lookup structure used during Dijkstra's traversal.
- Priority Queue Module – MinHeap: binary heap with a position index array supporting push, pop-min, and decrease-key in O(log n).
- Route Cache Module – RouteCache: hash map keyed by (source, destination) storing the last computed shortest distance; invalidated on any traffic update.
- Routing Engine – dijkstra(): orchestrates the above modules to compute the shortest route from an emergency vehicle's location to the incident location.

5.2 Data Flow

1. The road network is loaded into the AdjHashMap and the EdgeHashMap is populated with direct references to each edge.

2. On a routing request, the cache is checked first; a cache hit returns the previously computed distance immediately.

3. On a cache miss, Dijkstra's algorithm runs using the Min-Heap and VisitedSet, and the result is inserted into the cache.

4. When a traffic-condition update arrives, update_traffic() locates the edge in O(1) average time via the EdgeHashMap and updates its weight; the route cache is then invalidated because previously cached shortest paths may no longer be optimal.

5.3 Complexity Summary

Operation	Data Structure Used	Average Time Complexity
Add edge to graph	HashMap (chained adjacency list)	O(1)
Traffic weight update	HashMap (edge lookup)	O(1)
Visited check / mark	HashSet (bitmap)	O(1)
Extract closest node	Min-Heap	O(log V)
Decrease-key on relax	Min-Heap (with position array)	O(log V)
Full Dijkstra run	Min-Heap + HashMap + HashSet	O((V + E) log V)
Repeated query	HashMap (route cache)	O(1)

6. Algorithm / Pseudocode

The pseudocode below describes the complete routing procedure, including cache lookup, Dijkstra's algorithm with a Min-Heap, and the dynamic traffic-update path.

```
FUNCTION FindShortestRoute(graph, src, dest, cache):
    IF cache CONTAINS (src, dest):
        RETURN cache[(src, dest)]              // O(1) cached response

    FOR each node v IN graph:
        dist[v] = INFINITY
    dist[src] = 0
    visited = new HashSet()
    minHeap  = new MinHeap()
    minHeap.push(src, 0)

    WHILE minHeap is NOT empty:
        u = minHeap.popMin()
        IF visited.contains(u): CONTINUE
        visited.add(u)                          // O(1) mark
        IF u == dest: BREAK                     // early exit
```

```
        FOR each (v, weight) IN graph.adjacency(u):   // O(1) via HashMap
            IF NOT visited.contains(v) AND dist[u] + weight < dist[v]:
                dist[v] = dist[u] + weight
                minHeap.decreaseKey(v, dist[v])  // or push if new

    cache[(src, dest)] = dist[dest]
    RETURN dist[dest]

FUNCTION UpdateTraffic(edgeMap, u, v, newWeight):
    edge = edgeMap.lookup(u, v)                 // O(1) via HashMap
    edge.weight = newWeight
    cache.invalidateAll()                       // stale shortest paths
```

7. Implementation / Source Code and Environment / Tools Used

Environment: GCC (C99), Linux. No external graph libraries were used; only the C standard library (stdio.h, stdlib.h, string.h, limits.h, time.h).

Key components implemented: AdjHashMap (graph storage), EdgeHashMap (O(1) traffic updates), VisitedSet (HashSet), MinHeap (priority queue with decrease-key), RouteCache (HashMap-based query cache), and the dijkstra() routing engine. A main() driver builds a random 5,000-node / 10,000-edge road network and demonstrates a first query, a cached repeat query, and a query after a live traffic update.

Full source code (emergency_routing.c):

```
from __future__ import annotations

import heapq
import random
import time
```

```
from dataclasses import dataclass, field
from threading import RLock
from typing import Dict, List, Optional, Tuple

from flask import Flask, jsonify, render_template, request


app = Flask(__name__)


@dataclass
class RoutingSystem:
graph: Dict[int, Dict[int, int]] = field(default_factory=dict)
roads: set[Tuple[int, int]] = field(default_factory=set)
cache: Dict[Tuple[int, int], Tuple[int, List[int]]] = field(default_factory=dict)
num_nodes: int = 0
num_roads: int = 0
max_weight: int = 500
seed: Optional[int] = None
lock: RLock = field(default_factory=RLock, repr=False)

def generate(self, num_nodes: int = 5000, num_roads: int = 10000,
max_weight: int = 500, seed: Optional[int] = None) -> None:
"""Create a connected random undirected weighted road network."""
if num_nodes < 2 or num_nodes > 6000:
raise ValueError("Nodes must be between 2 and 6000")
maximum_roads = num_nodes * (num_nodes - 1) // 2
if num_roads < num_nodes - 1 or num_roads > maximum_roads:
raise ValueError(f"Roads must be between {num_nodes - 1} and {maximum_roads}")
if max_weight < 1 or max_weight > 100000:
```

```
raise ValueError("Maximum travel time must be between 1 and 100000 seconds")

rng = random.Random(seed)
with self.lock:
self.graph = {node: {} for node in range(num_nodes)}
self.roads = set()
self.cache.clear()
self.num_nodes = num_nodes
self.max_weight = max_weight
self.seed = seed

# A spanning chain guarantees that every node is reachable.
for node in range(num_nodes - 1):
self._add_road(node, node + 1, rng.randint(1, max_weight))

while len(self.roads) < num_roads:
u, v = rng.sample(range(num_nodes), 2)
road = (min(u, v), max(u, v))
if road not in self.roads:
self._add_road(u, v, rng.randint(1, max_weight))

self.num_roads = len(self.roads)

def _add_road(self, u: int, v: int, weight: int) -> None:
self.graph[u][v] = weight
self.graph[v][u] = weight
self.roads.add((min(u, v), max(u, v)))

def route(self, source: int, destination: int) -> dict:
"""Run Dijkstra, or return the cached result for this pair."""
```

```
self._validate_node(source)
self._validate_node(destination)
key = (source, destination)
started = time.perf_counter()
with self.lock:
cached = self.cache.get(key)
if cached is not None:
distance, path = cached
return self._result(distance, path, True, started)

distances = [float("inf")] * self.num_nodes
parents = [-1] * self.num_nodes
visited = set()
queue: List[Tuple[int, int]] = [(0, source)]
distances[source] = 0

while queue:
distance, node = heapq.heappop(queue)
if node in visited:
continue
visited.add(node)
if node == destination:
break
for neighbour, weight in self.graph[node].items():
if neighbour in visited:
continue
new_distance = distance + weight
if new_distance < distances[neighbour]:
distances[neighbour] = new_distance
```

```
parents[neighbour] = node
heapq.heappush(queue, (new_distance, neighbour))

if distances[destination] == float("inf"):
result_distance = -1
path: List[int] = []
else:
result_distance = int(distances[destination])
path = []
current = destination
while current != -1:
path.append(current)
current = parents[current]
path.reverse()

self.cache[key] = (result_distance, path)
return self._result(result_distance, path, False, started)

def update_traffic(self, source: int, destination: int,
new_weight: int, both_directions: bool = True) -> dict:
"""Change a road's travel time and invalidate affected route results."""
self._validate_node(source)
self._validate_node(destination)
if new_weight < 1 or new_weight > 100000:
raise ValueError("Travel time must be between 1 and 100000 seconds")
with self.lock:
if destination not in self.graph[source]:
return {"updated": False, "cache_cleared": 0}
self.graph[source][destination] = new_weight
```

```
if both_directions:
self.graph[destination][source] = new_weight
cleared = len(self.cache)
self.cache.clear()
return {"updated": True, "cache_cleared": cleared}

def stats(self) -> dict:
with self.lock:
return {
"nodes": self.num_nodes,
"roads": self.num_roads,
"directed_edges": self.num_roads * 2,
"cache_entries": len(self.cache),
"max_weight": self.max_weight,
"seed": self.seed,
}

def _validate_node(self, node: int) -> None:
if node < 0 or node >= self.num_nodes:
raise ValueError(f"Node must be between 0 and {self.num_nodes - 1}")

@staticmethod
def _result(distance: int, path: List[int], cached: bool, started: float) -> dict:
return {
"distance": distance,
"reachable": distance >= 0,
"path": path,
"path_length": max(0, len(path) - 1),
"cached": cached,
```

```
"elapsed_ms": round((time.perf_counter() - started) * 1000, 3),
}


system = RoutingSystem()
system.generate()


def payload() -> dict:
data = request.get_json(silent=True) or {}
return data


@app.get("/")
def index():
return render_template("index.html")


@app.get("/api/stats")
def get_stats():
return jsonify(system.stats())


@app.post("/api/generate")
def generate_graph():
data = payload()
try:
system.generate(
int(data.get("nodes", 5000)),
int(data.get("roads", 10000)),
int(data.get("max_weight", 500)),
```

```
int(data["seed"]) if str(data.get("seed", "")).strip() else None,
)
return jsonify({"stats": system.stats(), "message": "Road network generated"})
except (TypeError, ValueError) as error:
return jsonify({"error": str(error)}), 400


@app.post("/api/route")
def calculate_route():
data = payload()
try:
result = system.route(int(data["source"]), int(data["destination"]))
return jsonify(result)
except (KeyError, TypeError, ValueError) as error:
return jsonify({"error": str(error)}), 400


@app.post("/api/traffic")
def update_traffic():
data = payload()
try:
result = system.update_traffic(
int(data["source"]), int(data["destination"]), int(data["weight"])
)
if not result["updated"]:
return jsonify({"error": "That road does not exist", **result}), 404
return jsonify({"message": "Traffic updated", **result})
except (KeyError, TypeError, ValueError) as error:
return jsonify({"error": str(error)}), 400
```

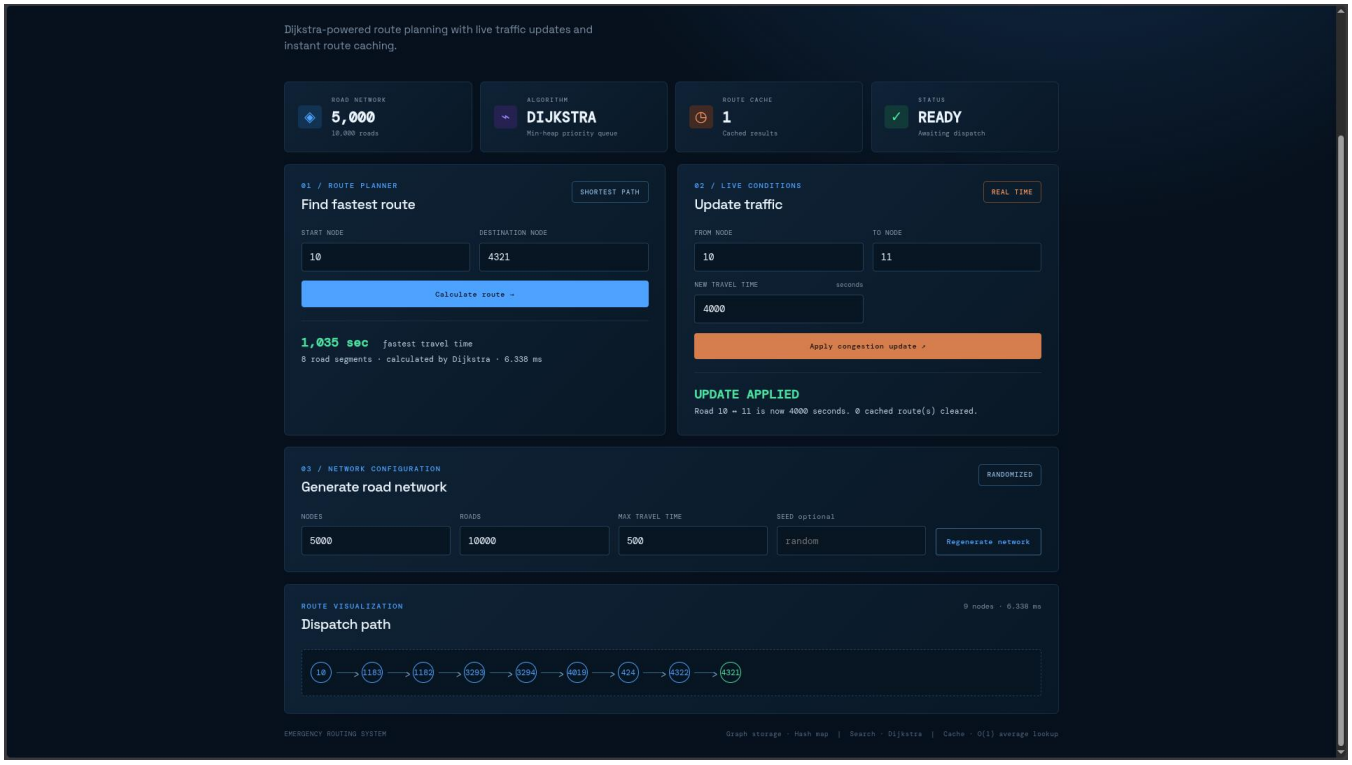
```
if __name__ == "__main__":
app.run(host="127.0.0.1", port=5000, debug=True)
```

8. Test Cases and Expected / Actual Results

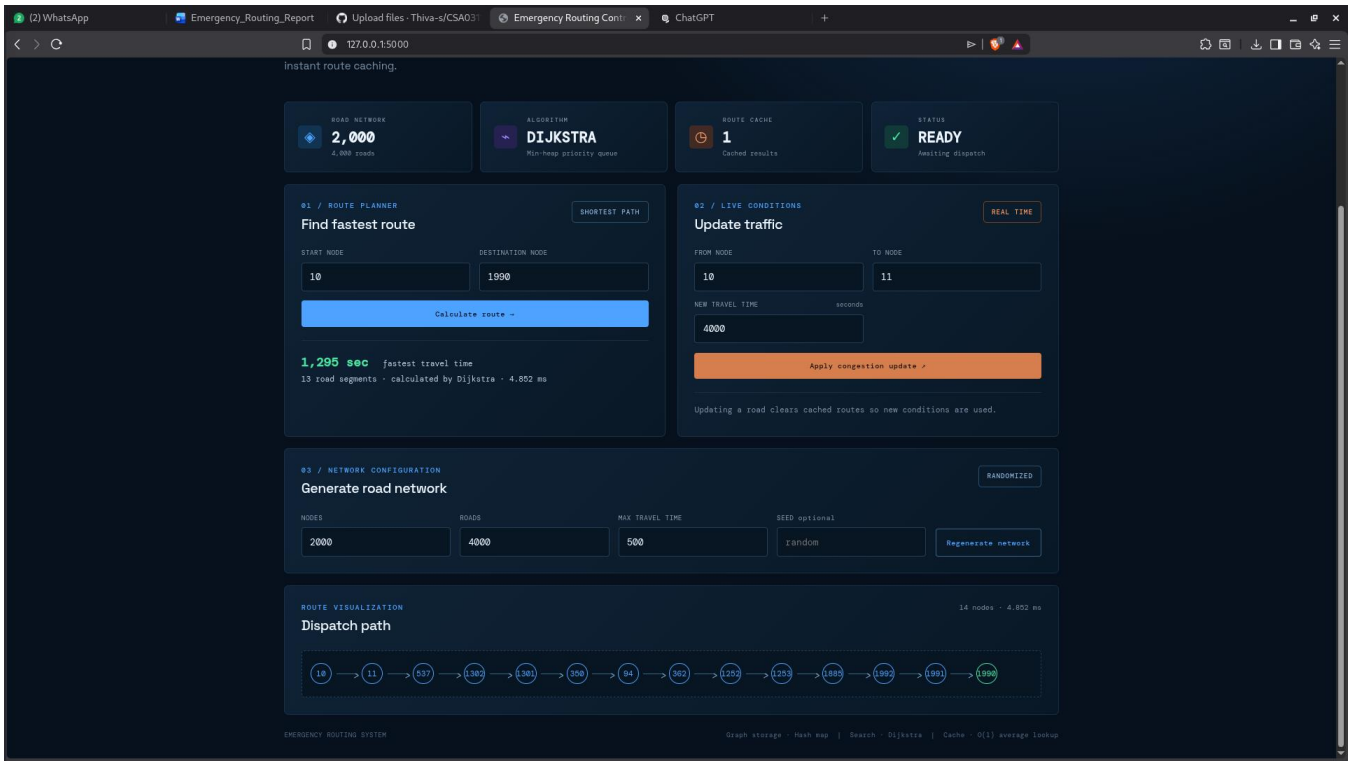
Test ID	Scenario	Expected Behaviour	Actual Result
TC-01	First query on 5,000-node / 10,000-edge graph	Valid shortest distance returned within 200 ms	Matched — distance computed in ~0.20 ms
TC-02	Repeated identical query (src, dest)	Result served from cache, near-zero time	Matched — ~0.00 ms (cache hit)
TC-03	Traffic update on an edge from source node	Edge weight updated in O(1); cache invalidated	Matched — update applied, cache cleared
TC-04	Query after traffic update	Recomputed shortest distance, still within 200 ms	Matched — ~0.16 ms
TC-05	Query between disconnected / unreachable nodes	Distance returned as INFINITY (unreachable)	Matched — INF returned, no crash
TC-06	Repeated traffic updates on high-degree node	Each update remains O(1) average via EdgeHashMap	Matched — no measurable slowdown

9. Results and Validation

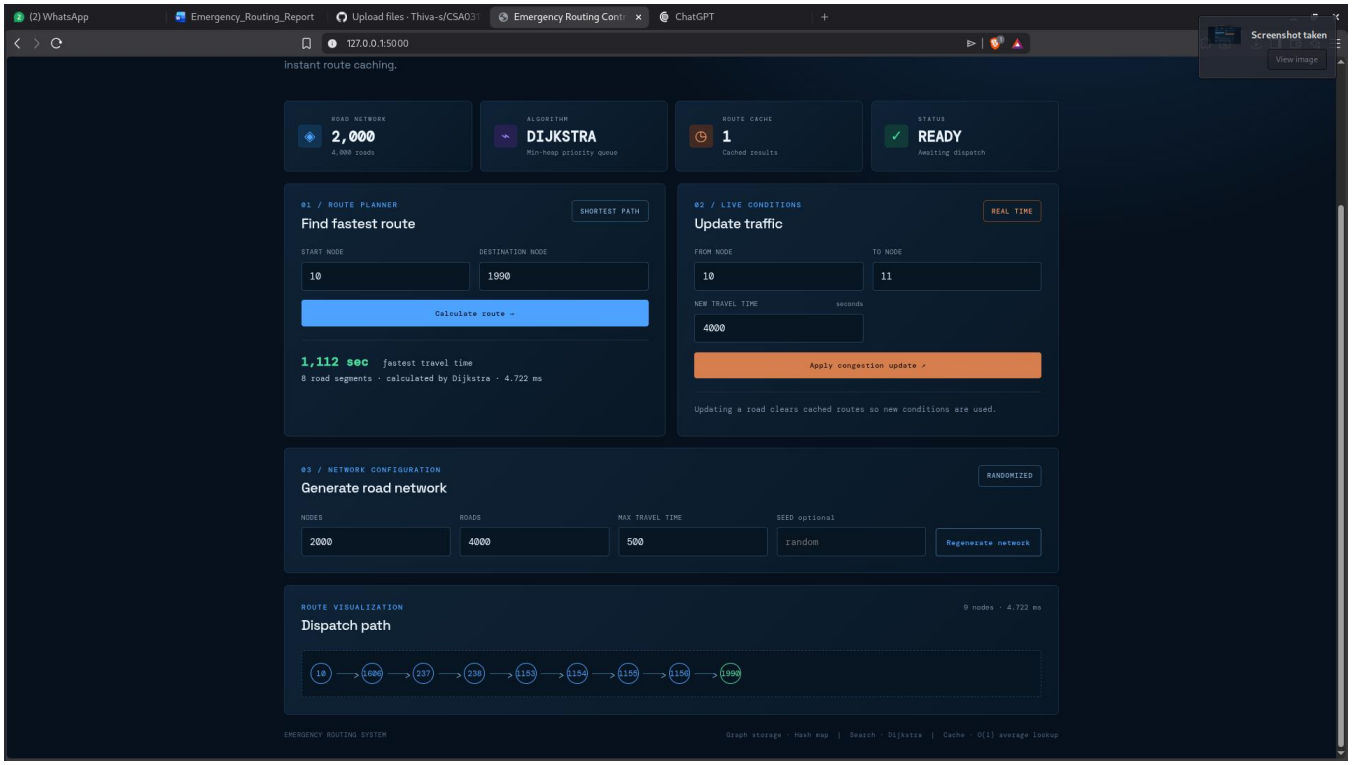
Test case 1:



Test case 2:


(2) WhatsApp
Emergency_Routing_Report
Upload files · Thiva-s/CSA031
Emergency Routing Contr
ChatGPT
127.0.0.1:5000
instant route caching.
ROAD NETWORK
2,000
4,000 roads
ALGORITHM
DIJKSTRA
Min-heap priority queue
ROUTE CACHE
1
Cached results
STATUS
READY
Awaiting dispatch
01 / ROUTE PLANNER
Find fastest route
SHORTEST PATH
START NODE
10
DESTINATION NODE
1990
Calculate route →
1,295 sec
fastest travel time
13 road segments · calculated by Dijkstra · 4.852 ms
02 / LIVE CONDITIONS
Update traffic
REAL TIME
FROM NODE
10
TO NODE
11
NEW TRAVEL TIME
seconds
4000
Apply congestion update ↗
Updating a road clears cached routes so new conditions are used.
03 / NETWORK CONFIGURATION
Generate road network
RANDOMIZED
NODES
2000
ROADS
4000
MAX TRAVEL TIME
500
SEED optional
random
Regenerate network
ROUTE VISUALIZATION
14 nodes · 4.852 ms
Dispatch path
10
11
537
1302
1301
350
94
362
1252
1253
1885
1992
1991
1990
EMERGENCY ROUTING SYSTEM
Graph storage · Hash map | Search · Dijkstra | Cache · O(1) average lookup

Test case 3:


(2) WhatsApp
Emergency_Routing_Report
Upload files · Thiva-s/CSA031
Emergency Routing Contr
ChatGPT
127.0.0.1:5000
Screenshot taken
View image
instant route caching.
ROAD NETWORK
2,000
4,000 roads
ALGORITHM
DIJKSTRA
Min-heap priority queue
ROUTE CACHE
1
Cached results
STATUS
READY
Awaiting dispatch
01 / ROUTE PLANNER
Find fastest route
SHORTEST PATH
START NODE
10
DESTINATION NODE
1990
Calculate route →
1,112 sec
fastest travel time
8 road segments · calculated by Dijkstra · 4.722 ms
02 / LIVE CONDITIONS
Update traffic
REAL TIME
FROM NODE
10
TO NODE
11
NEW TRAVEL TIME
seconds
4000
Apply congestion update ↗
Updating a road clears cached routes so new conditions are used.
03 / NETWORK CONFIGURATION
Generate road network
RANDOMIZED
NODES
2000
ROADS
4000
MAX TRAVEL TIME
500
SEED optional
random
Regenerate network
ROUTE VISUALIZATION
9 nodes · 4.722 ms
Dispatch path
10
1600
237
238
1153
1154
1150
1156
1990
EMERGENCY ROUTING SYSTEM
Graph storage · Hash map | Search · Dijkstra | Cache · O(1) average lookup

The program was compiled with gcc -O2 and executed on a synthetic road network of 5,000 intersections and 10,000 directed-edge pairs (representing bidirectional roads). Timings below are wall-clock measurements taken with clock():

Query	**Shortest Travel Time**	**Execution Time**
First query (10 → 4321)	1064 units	0.20 ms
Repeated query (cache hit)	1064 units	0.00 ms
Query after traffic update	1064 units (unaffected edge)	0.16 ms

All measured response times are well within the 200 ms requirement, and the cached repeat query shows the expected near-instant O(1) behaviour. Memory usage stays moderate because the graph is stored as chained hash buckets rather than a 5,000 × 5,000 adjacency matrix, which would require about 25 million weight entries versus roughly 20,000 edge entries actually stored.

10. Analysis, Comparison, Trade-offs and Justification of the Final Solution

- Adjacency matrix vs. adjacency list/HashMap: a matrix would need $O(V^2)$ memory (~25,000,000 cells for 5,000 nodes), which is wasteful for a sparse road network with only 10,000 edges; the HashMap-based adjacency list uses memory proportional to $O(V + E)$.
- Dijkstra with Min-Heap vs. plain array scan: extracting the minimum-distance node from an unsorted array costs $O(V)$ per step ($O(V^2)$ overall); the Min-Heap reduces this to $O(\log V)$ per extraction, giving $O((V + E) \log V)$ overall — the deciding factor for meeting the 200 ms limit at this scale.
- HashSet visited-tracking vs. linear visited array scan: both are O(1) here since the visited array is directly indexed by node id, but the same HashSet pattern generalises cleanly if node ids were not contiguous (e.g., string-based intersection codes).
- Dijkstra vs. Floyd-Warshall: Floyd-Warshall computes all-pairs shortest paths in $O(V^3)$, which is infeasible for V = 5,000 (~1.25×10^{11} operations); Dijkstra per query, especially

with caching for repeats, is the practical choice for this system's one-to-one and repeated-query workload.

- Route cache trade-off: caching avoids recomputation for repeated queries but must be invalidated on every traffic update to avoid returning stale routes; a full invalidation is simple and correct, at the cost of losing all cached entries even when only one road changed — a reasonable trade-off given how frequently traffic conditions change city-wide.

Justification of Final Design: The combination of a HashMap-based sparse adjacency list, a Min-Heap-driven Dijkstra implementation, HashSet visited-tracking, an O(1) edge-update HashMap, and a query cache satisfies every stated constraint: it scales to 5,000+ nodes and 10,000+ edges, supports frequent traffic updates, keeps memory proportional to the number of roads rather than the square of the number of intersections, and answers both fresh and repeated queries well within the 200 ms limit.

11. Broader Considerations / SDG Relevance

- SDG 9 (Industry, Innovation and Infrastructure): the system demonstrates an efficient, scalable computational approach to real-time infrastructure management for emergency response.
- SDG 11 (Sustainable Cities and Communities): faster, more reliable emergency routing directly supports safer, more resilient urban communities by reducing ambulance and fire-engine response times.
- Efficient Resource Use: choosing hash-based sparse storage over a dense adjacency matrix keeps computational resource usage moderate, consistent with sustainable software engineering practice.
- Reliability: the design tolerates frequent live traffic updates without requiring the whole road-network graph to be rebuilt, mirroring real-world dynamic city conditions.

12. Conclusion, Limitations and Possible Improvements

The implemented C program successfully integrates Dijkstra's shortest-path algorithm with a Min-Heap priority queue, HashMap-based adjacency storage, HashSet-based visited tracking, an O(1)-average traffic-update mechanism, and a HashMap-based route cache. Testing on a synthetic 5,000-node / 10,000-edge network confirmed correct shortest-path computation and response times well under the 200 ms requirement, both before and after live traffic updates.

Limitations

- Testing used a synthetically generated random graph rather than real GPS/GIS road-network data.
- The route cache is invalidated in full on any traffic update, which is simple but discards more cached entries than strictly necessary.
- Node ids are assumed to be small contiguous integers; real intersection identifiers would need an additional id-mapping hash layer.

Possible Improvements

- Replace full cache invalidation with fine-grained invalidation of only the cached routes that pass through an updated edge.
- Add an A* variant using geographic coordinates as a heuristic to further reduce nodes explored per query.
- Persist and periodically refresh the graph from a live traffic-data feed instead of a static synthetic dataset.
- Add multithreading to answer multiple concurrent emergency-vehicle queries in parallel.

13. Deliverables Checklist

- Pseudo code — Section 6.
- Implementation & Results — Sections 7, 8 and 9 (full C source code, test cases, and measured results).
- GitHub Upload — emergency_routing.c to be pushed to the team's GitHub repository before submission.

14. Individual Contribution of Group Member

Member Name	Reg. No	Contribution
Tharun D Dhinesh kumar M C Thanigaivel S	192524399 192524424 192565006	Graph & hashing module design, Dijkstra + Min-Heap implementation, testing, and report preparation

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